

David J. Halpern

djh2142@columbia.edu

Zuckerman Mind, Brain, Behavior Institute

Columbia University

3227 Broadway

New York, NY, 10027

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Employment History

- 2025 – **Associate Research Scientist.** Zuckerman Mind, Brain, Behavior Institute, Columbia University
Advisor: Daphna Shohamy
- 2020 – 2024 **Postdoctoral Fellow.** Department of Psychology, University of Pennsylvania
Advisor: Michael Kahana

Education

- 2020 **Ph.D.**, Psychology (Cognition and Perception), New York University
Advisor: Todd Gureckis
- 2019 **M.A.**, Psychology, New York University
- 2013 **B.A.** Psychology and Music, Columbia University

Preprints and Forthcoming

Halpern, D. J., & Gureckis, T. (In Revision). Getting blood from a stone: Improving neural inferences without neural data. *Imaging Neuroscience*. <https://www.biorxiv.org/content/10.1101/2021.01.21.427334v2>

Jin, B., **Halpern, D. J.**, & Kahana, M. J. (In Revision). A theory of memory for items and associations. *Psychological Review*. <https://doi.org/10.31234/osf.io/xm8v6>

Rafla, D., **Halpern, D. J.**, Katerman, B. S., & Kahana, M. J. (In Revision). Organizational dynamics of memory across days. *Journal of Experimental Psychology: Learning, Memory and Cognition*.
<https://doi.org/10.31234/osf.io/j7gkd>

Peer-Reviewed Journal Articles

Halpern, D. J., Lega, B. C., Gross, R. E., Wu, C., Sperling, M. R., Aronson, J. P., Jobst, B. C., & Kahana, M. J. (2025). Study-phase reinstatement predicts subsequent recall. *Nature Neuroscience*, 28(4), 883–890.
<https://doi.org/10.1038/s41593-025-01884-8>

Rudoler, J. H., Bruska, J. P., Chang, W., Dougherty, M. R., Katerman, B. S., **Halpern, D. J.**, Diamond, N. B., & Kahana, M. J. (2024). Decoding eeg for optimizing naturalistic memory. *Journal of Neuroscience Methods*, 410, 110220. <https://doi.org/10.1016/j.jneumeth.2024.110220>

Sakon, J. J., **Halpern, D. J.**, Schonhaut, D. R., & Kahana, M. J. (2024). Human hippocampal ripples signal encoding of episodic memories. *The Journal of Neuroscience*, e0111232023.
<https://doi.org/10.1523/jneurosci.0111-23.2023>

Halpern, D. J., Tubridy, S., Davachi, L., & Gureckis, T. M. (2023). Identifying causal subsequent memory effects. *Proceedings of the National Academy of Sciences*, 120(13),
<https://www.pnas.org/doi/pdf/10.1073/pnas.2120288120>, e2120288120.
<https://doi.org/10.1073/pnas.2120288120>

Adroque, R. T., Herz, N., **Halpern, D. J.**, Tracy, J., & Kahana, M. J. (2023). Multitrial free recall for evaluating memory. *Neuropsychology*. <https://doi.org/10.1037/neu0000910>

- Dougherty, M. R., **Halpern, D. J.**, & Kahana, M. J. (2023). Forward and backward recall dynamics. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. <https://doi.org/10.1037/xlm0001254>
- van Baar, J. M., **Halpern, D. J.**, & FeldmanHall, O. (2021). Intolerance of uncertainty modulates brain-to-brain synchrony during politically polarized perception. *Proceedings of the National Academy of Sciences*, 118(20), <https://www.pnas.org/doi/pdf/10.1073/pnas.2022491118>, e2022491118. <https://doi.org/10.1073/pnas.2022491118>
- Nussenbaum, K., Cohen, A. O., Davis, Z. J., **Halpern, D. J.**, Gureckis, T. M., & Hartley, C. A. (2020). Causal information-seeking strategies change across childhood and adolescence. *Cognitive Science*, 44(9), e12888. <https://onlinelibrary.wiley.com/doi/full/10.1111/cogs.12888>
- Gureckis, T. M., Martin, J., McDonnell, J., Rich, A. S., Markant, D., Coenen, A., **Halpern, D. J.**, Hamrick, J. B., & Chan, P. (2016). psiTurk: An open-source framework for conducting replicable behavioral experiments online. *Behavior Research Methods*, 48(3), 829–842.

Peer-Reviewed Conference Proceedings

- Nussenbaum, K., Cohen, A. O., Davis, Z., **Halpern, D. J.**, Gureckis, T., & Hartley, C. (2019). Causal intervention strategies change across development, In *Proceedings of the 41st Annual Conference of the Cognitive Science Society*.
- Halpern, D. J.**, & Rodriguez, P. (2018). Partisan representations: Partisan differences in semantic representations and their role in attitude judgments., In *Proceedings of the 40th Annual Conference of the Cognitive Science Society*. https://cognitivesciencesociety.org/wp-content/uploads/2019/01/cogsci18_proceedings.pdf
- Halpern, D. J.**, Tubridy, S., Davachi, L., & Gureckis, T. (2018). Predicting memory performance using a joint model of brain and behavior, In *Proceedings of the 2nd Annual Computational Cognitive Neuroscience Conference*.
- Halpern, D. J.**, Tubridy, S., Wang, H., Gasser, C., Popp, P., Davachi, L., & Gureckis, T. (2018). Knowledge tracing using the brain, In *Proceedings of the 11th International Conference on Educational Data Mining*. https://educationaldatamining.org/files/conferences/EDM2018/papers/EDM2018_paper_158.pdf
- Rich, A., Popp, P. O., **Halpern, D. J.**, Rothe, A., & Gureckis, T. (2018). Modeling second-language learning from a psychological perspective, In *Proceedings of the 13th Workshop on Innovative Use of NLP for Building Educational Applications*.
- Tubridy, S., **Halpern, D. J.**, Davachi, L., & Gureckis, T. M. (2018). A neurocognitive model for predicting the fate of individual memories, In *Proceedings of the 40th Annual Conference of the Cognitive Science Society*. https://cognitivesciencesociety.org/wp-content/uploads/2019/01/cogsci18_proceedings.pdf
- Halpern, D. J.**, & Gureckis, T. M. (2017). Categorization, information selection and stimulus uncertainty., In *Proceedings of the 39th Annual Conference of the Cognitive Science Society*. https://cognitivesciencesociety.org/wp-content/uploads/2019/01/cogsci17_proceedings.pdf

Conference Abstracts

- Halpern, D. J.**, & DeHaan, R. (2024). Predicting effects of brain stimulation from observational data, In *31st Annual Cognitive Neuroscience Society Meeting*.
- Halpern, D. J.**, DeHaan, R., & Kahana, M. J. (2024). Predicting effects of brain stimulation from observational data, In *2024 Society for Neuroscience Meeting*.
- Halpern, D. J.**, & Kahana, M. J. (2024). Study-phase reinstatement: Encoding spontaneous thoughts as memories, In *20th Annual Context and Episodic Memory Symposium*.
- Halpern, D. J.**, & Kahana, M. J. (2023a). Covert reinstatement predicts recall probability, In *2023 Society for Neuroscience Meeting*.
- Halpern, D. J.**, & Kahana, M. J. (2023b). Covert reinstatement predicts recall probability and organization, In *2023 International Conference on Learning and Memory*.
- Dougherty, M., **Halpern, D. J.**, & Kahana, M. J. (2022). Forward and backward serial recall, In *Proceedings of the 44th Annual Conference of the Cognitive Science Society*.
- Halpern, D. J.**, Chang, W., & Kahana, M. J. (2022a). The role of memory search in evaluations, In *18th Annual Context and Episodic Memory Symposium*.

- Halpern, D. J.,** Chang, W., & Kahana, M. J. (2022b). The role of memory search in evaluations, In *55th Annual Meeting of the Society for Mathematical Psychology*.
- Halpern, D. J.,** & Kahana, M. J. (2022a). Covert reinstatement predicts recall initiation, In *19th Annual Computational and Systems Neuroscience (Cosyne) Meeting*.
- Halpern, D. J.,** & Kahana, M. J. (2022b). Covert reinstatement predicts recall initiation, In *29th Annual Cognitive Neuroscience Society Meeting*.
- Halpern, D. J.,** & Kahana, M. J. (2021). Directly measuring reactivation of memorized content directly measuring reactivation of memorized content with electrophysiology., In *17th Annual Context and Episodic Memory Symposium*.
- van Baar, J., **Halpern, D. J.,** Rodriguez, P., Hasson, U., & FeldmanHall, O. (2020). The multimodal roots of polarized political perception, In *13th Annual Meeting of the Social Affective Neuroscience Society*.
- Halpern, D. J.,** & Rodriguez, P. (2019). Similarity in social space predicts similarity of semantic space, In *52nd Annual Meeting of the Society for Mathematical Psychology*.
- Halpern, D. J.,** Tubridy, S., Davachi, L., & Gureckis, T. (2019). Combining neural and behavioral data to predict human long-term memory, In *52nd Annual Meeting of the Society for Mathematical Psychology*.
- Rodriguez, P., & **Halpern, D. J.** (2018a). Exploring the role of semantic memory in attitude judgments, In *114th American Political Science Association Annual Meeting*.
- Rodriguez, P., & **Halpern, D. J.** (2018b). Political attitudes: Exploring the role of semantic memory in attitude judgments, In *41st Annual Meeting of the International Society for Political Psychology*.
- Halpern, D. J.,** & Rodriguez, P. (2017). Beyond semantics: Semantic memory influences attitude judgments, In *58th Annual Meeting of the Psychonomic Society*.
- Tubridy, S., **Halpern, D. J.,** Davachi, L., & Gureckis, T. (2017). A hierarchical bayesian approach to inferring mnemonic status from the brain, In *1st Annual Computational Cognitive Neuroscience Conference*.
- Halpern, D. J.,** & Gureckis, T. (2016). Constrained optimization of measurements: A model of information selection in categorization, In *49th Annual Meeting of the Society for Mathematical Psychology*.
- Halpern, D. J.,** Howell, B., & Keller, R. (2013). Applications of music analysis algorithms to interactive music, In *The Improvising Brain Symposium*.

Grants

- (2023-2028) National Institutes of Health, R01MH055687
 "Associative Processes in Episodic Memory"
 \$366,466 annual direct costs.
 Key Contributor to Writing, Design and Implementation

Fellowships and Awards

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|-------------|--|
| Summer 2018 | Computational Cognitive Neuroscience Conference Travel Grant |
| Fall 2017 | NYU Dean's Travel Award |
| Summer 2016 | Richard Shiffrin Travel Award |
| Summer 2015 | Okinawa Computational Neuroscience Course |
| Summer 2012 | Harvey Mudd NSF REU in Systems |
| Summer 2011 | Amgen Scholars Summer Research Fellow |